AeroRIT: A New Scene for Hyperspectral Image Analysis

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***Abstract*— We investigate applying convolutional neural network (CNN) architecture to facilitate aerial hyperspectral scene understanding and present a new hyperspectral data set, AeroRIT, which is large enough for CNN training. To date, the majority of hyperspectral airborne has been confined to various subcategories of vegetation and roads, and this scene introduces two new categories: buildings and cars. To the best of our knowledge, this is the first comprehensive large-scale hyperspectral scene with nearly seven-million pixel annotations for identifying cars, roads, and buildings. We compare the performance of the three popular architectures—SegNet, U-Net, and Res-U-Net, for scene understanding and object identification via the task of dense semantic segmentation to establish a benchmark for the scene. To further strengthen the network, we add squeeze and excitation blocks for better channel interactions and use self-supervised learning for better encoder initialization. Aerial hyperspectral image analysis has been restricted to small data sets with limited train/test splits capabilities, and we believe that *AeroRIT* will help advance the research in the field with a more complex object distribution to perform well on. The full data set, with flight lines in radiance and reflectance domains, is available for download at https://github.com/aneesh3108/AeroRIT. This data set is the first step toward developing robust algorithms for hyperspectral airborne sensing that can robustly perform advanced tasks such as vehicle tracking and occlusion handling.**

***Index Terms*— hyperspectral imaging, image segmentation, supervised learning.**

I. Introduction

CONVOLUTIONAL neural networks (CNNs) are now being widely used for analyzing remote sensing imagery and though they have achieved some success, even the most well-designed CNNs for RGB imagery struggle to achieve a mean intersection-over-union of more than 80% on the ISPRS aerial data sets Vaihengen[1] and Potsdam[2] [1], [2]. This performance is in spite of the fact that these data sets have significantly higher spatial resolution with approximate ground sampling distances (GSDs) of 9 cm (Vaihengen) and 5 cm (Potsdam). One potential way to develop a better classifier is to include more discriminative signatures by moving from the RGB domain to the finer spectral resolution of hyperspectral imaging (HSI) systems. HSI systems record a contiguous spectrum, usually in steps of 1 or 5 nanometers (nm), which details the contents present in the scene and can assist in increasing discrimination capability. Despite the potential for improved spectral features, HSI data have been largely unused in machine learning applications. Partially, this is because HSI sensors are significantly more expensive than their RGB counterparts, thus leading to HSI data being restricted to domains such as precision agriculture and environmental monitoring. The largest popular data set in hyperspectral remote sensing for scene understanding—University of Pavia—has a spatial resolution of only 610 × 340, out of which nearly 80% samples are in the *undefined* class. The lack of data has made developing and training a CNN to leverage the addition spectral features of HSI prohibitively difficult. In this article, we seek to extend CNN-based architectures developed for medical domain [3] and RGB domain [4], [5] to use the additional spectral signatures of HSI data. Training such networks requires a lot of HSI data, so we introduce AeroRIT, a data set nearly eight times larger than the University of Pavia and with only 17% pixels under the undefined class category. While it is possible to analyze the data on a per-pixel basis, the wide variety of object distribution calls for a more structure-aware approach, and hence, we adopt semantic segmentation as the task of interest for this article.

There has been an interest in using CNNs for analyzing remote sensing imagery [6]–[10]. Uzkent *et al.* [6] adapted correlation filters trained on RGB images with HSI bands to successfully track cars in moving platform scenarios. Hughes *et al.* [7] used a siamese CNN architecture to match high-resolution optical images with their corresponding synthetic aperture radar images. Kleynhans *et al.* [8] compared the performance of predicting top-of-atmosphere thermal radiance by using forward modeling with radiosonde data and

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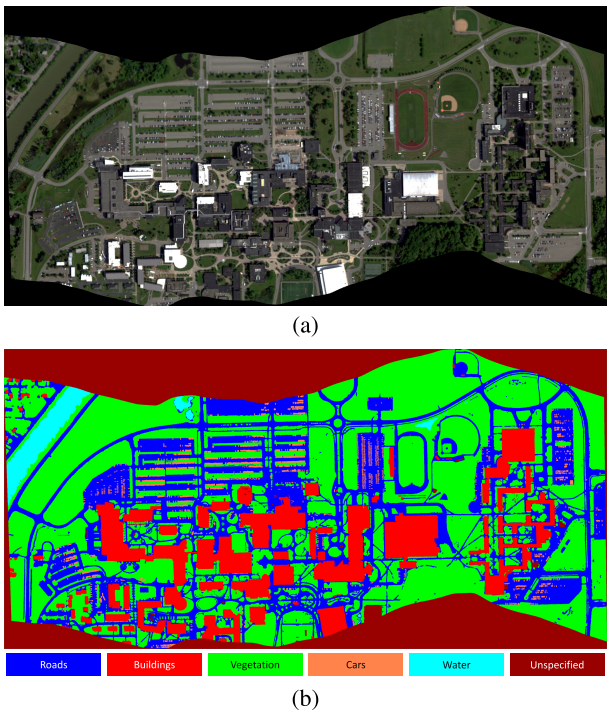
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radiative transfer modeling (MODTRAN) [11] against a multilayer perception (MLP) and CNN and observed better performance from MLP and CNN in all experimental cases. Kemker and Kanan [9] and Kemker *et al.* [10] used multiscale independent component analysis and stacked convolutional autoencoders as self-supervised learning tasks before performing semantic segmentation on multispectral and hyperspectral imagery.

The top three hyperspectral remote sensing data sets, Indian Pines, Salinas, and University of Pavia, have nearly distinctive classes, and hence, learning a discriminative boundary is relatively easier without the need for advanced architectures. One of the primary reasons for lack of HSI data sets is the cost associated with its collection—the costs of hardware and flight time are very expensive, and the data collection itself is weather-dependent. Assuming the costs can be offset by justifying the requirements of the task, we are also faced with high variance in spectral signatures overlooking the same scene due to factors such as atmospheric scattering and cloud cover. Furthermore, HSI sensors have varying filters (or spectral response curves), which make sensor configuration necessary metadata during all operations. Finally, based on the GSD of the scene, nonnadir RGB-trained CNN features may not provide highly discriminative information as spectral information could be lost by directly downsampling the channels using techniques such as principal component analysis (PCA) to RGB color dimension space.

To test the discriminative potential for spectral data in a more difficult setting, we flew an aircraft equipped with an HSI system and obtained multiple flight lines at different time stamps. We chose the flight line with the best combination of spatial and spectral quality and annotated every pixel within the flight line—we named the collect "AeroRIT" [Fig. 1(a)]. We focus on being able to distinguish between five classes: 1) roads; 2) buildings; 3) vegetation; 4) cars; and 5) water. This is the first data set having challenging endmember as the signatures of some classes (buildings and cars) tend to have a large manifold, and hence, generalization becomes tougher.

Mixed spectra, where multiple materials can present in single-pixel subject to GSD, are particularly challenging in remote sensing imagery due to their varying nature of the occurrence. Various spectral unmixing methods (survey: [12]) have been applied to separate mixed pixels, but most assume the composition of all elements in the scene, referred to as end-members, are previously known. However, it is impossible to have all information about end-members when the scene is constantly changing. For example, in a moving camera setup with a push-broom sensor, each scene typically contains multiple-colored cars and buildings, and applying spectral unmixing becomes difficult if the end-member signatures cannot be predetermined. We do not consider pseudo-end-members for the scope of this article, and we do not tackle spectral unmixing as a problem but address mixed pixels as sensor-level noise in our tasks. This is important as the goal of this article is to understand challenges in a moving camera setup, and although the scene is constant, we imagine the scene as one of the many time steps captured from an airborne system.



(a)

(b)

Fig. 1. AeroRIT scene overlooking Rochester Institute of Technology's university campus. The spatial resolution is 1973×3975 pixels and covers the spectral range of 397 to 1003 nm in 1-nm steps. (a) RGB rendered version of the scene. (b) Semantic labeling for the scene.

II. Related Work

A. Data Sets for Hyperspectral Remote Sensing Imagery

Table I briefly reviews the current extent of aerial hyperspectral data sets available for analysis. Other hyperspectral data sets include ICVL [13] and CAVE [14]; however, we do not include them in the table as they are nonnadir and do not have pixelwise labels for the data. The most commonly used aerial data sets are: 1) Indian Pines; 2) Salinas Valley; and 3) University of Pavia. The first two primarily contain vegetation, and the third contains classes typically found around a university, for example, trees, soil, and asphalt. In all the three cases, the small spatial extent often leads researchers to use Monte Carlo (MC) cross-validation splits for benchmarking the performance of various CNN-based architectures. Recently, Nalepa *et al.* [15] showed that MC splits can often lead to near-perfect results as there tends to be pixel overlap (leakage) between the training and test sets. This article also introduces a new routine that ensures minimum to no leakage between generated data splits. However, there is still a possibility of the network overfitting on the training set as the number of samples is significantly small (Table I). In our scene, we label every pixel of the flight line and create an overall hyperspectral data set package that contains the radiance image, reflectance image, and the semantic label for every pixel. We also provide a training, validation, and test set that can be used to benchmark performance without the need

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TABLE I

POPULAR BENCHMARK HSI DATA SETS USED FOR SEMANTIC SEGMENTATION (OR PIXEL CLASSIFICATION), WITH INFORMATION ON THE SPATIAL AND SPECTRAL RESOLUTION. OUR DATA SET IS HIGHLIGHTED AND, AS OBSERVED, IS SIGNIFICANTLY BIGGER THAN ITS COUNTERPARTS. (ACRONYMS: AVIRIS: AIRBORNE VISIBLE/INFRARED IMAGING SPECTROMETER, ROSIS: REFLECTIVE OPTICS SYSTEM IMAGING SPECTROMETER, AND HYDICE: HYPERSPECTRAL DIGITAL IMAGERY COLLECTION EXPERIMENT)

Dataset	Sensor	Spatial Dimensions [px]	Spectral Dimensions [nm]	Spectral Bands	No. of classes
Indian Pines	AVIRIS	145 × 145	400 - 2500	224	16
Salinas Valley	AVIRIS	512 × 217	400 - 2500	224	16
Univ. of Pavia	ROSIS	610 × 340	430 - 838	103	9
KSC	AVIRIS	512 × 614	400 - 2500	224	13
Samson	-	952 × 952	401 - 889	156	3
Jasper Ridge	AVIRIS	512 × 614	380 - 2500	224	4
Urban	HYDICE	307 × 307	400 - 2500	210	6
AeroRIT	**Headwall Micro E**	**1973 × 3975**	**397 - 1003**	**372**	**5**

for cross-validation splits. We describe the data collection for our scene in Section III.

B. Semantic Segmentation

Semantic segmentation in HSI is often treated as a pixel classification problem due to a lack of sufficient samples. Most approaches fall under three categories: 1) spectral classifiers; 2) spatial classifiers; and 3) spectral–spatial classifiers. Hu *et al.* [16] used 1D-CNNs to extract the spectral features of HSIs and establish a baseline. The 1D-CNN takes a pixel spectral vector as an input, followed by a convolution layer and a max-pooling layer to compute a final class label. Li *et al.* [17] proposed to extract pixel-pair features and treat classification as a Siamese network problem. Hao *et al.* [18] designed a two-stream architecture, where *stream1* used a stacked denoising autoencoder to encode the spectral values of each input pixel of a patch and *stream2* used a CNN to process the patch's spatial features. Zhu *et al.* [19] used a generative adversarial network (GAN) to create robust classifiers of hyperspectral signatures. Recently, Roy *et al.* [20] proposed using a 3D-CNN followed by a 2D-CNN to learn better abstract-level representations for HSI scenes. We refer readers to Li *et al.* [21] for an in-depth overview of recent methods for HSI classification. As the abovementioned methods do not perform semantic segmentation in the truest sense (classification: encoder → class label and segmentation: encoder → decoder), we do not include them in our network comparisons.

III. AERORIT

The AeroRIT scene was captured by flying two types of camera systems over the Rochester Institute of Technology's university campus in a Cessna aircraft. The first camera system consisted of an 80 Mpixel (MP), RGB, framing-type silicon sensor, while the second system consisted of a visible near-infrared (VNIR) hyperspectral Headwall Photonics Micro Hyperspec E-Series CMOS sensor. The entire data collection took place over a couple of hours, where the sky was completely free of cloud cover, except the last few flight lines at the end of the day where there was some sparse cloud cover. The aircraft was flown over the campus at an altitude of approximately 5000 feet, thus yielding an effective GSD of about 0.4 m for the hyperspectral imagery. The RGB data were orthorectified onto the Shuttle Radar Topography Mission (SRTM) v4.1 digital elevation model (DEM) while the HSI was rectified onto a flat plane at the average terrain height of the flight line (that is, a low-resolution DEM). Both data sets were calibrated to spectral radiance in units of $\mathrm{Wm^{-2}sr^{-1}}\mu\mathrm{m^{-1}}$. The pixels were labeled with ENVI,[3] using individual hyperspectral signatures and the georegistered RGB images as references. As the RGB images do not form a continuous flight line (framing camera pattern) and are more in short burst captures format, we only labeled the hyperspectral scene and use it in our analysis.

Some important challenges associated with the scene are as follows.

1) *Low-Resolution:* CNNs have been known to learn edge- and color-related features in the early- to mid-layers [22]. In our case, the low pixel resolution coupled with mixed pixels makes discriminative feature learning relatively difficult [Fig. 2(a) and (b)].
2) *Glint:* Sun glint occurs due to bidirectional reflectance and the surface paint directly reflecting sunlight into the camera sensor. We observe that this only occurs in certain parts of the imagery and is almost always associated with vehicles. As identifying pixels of the vehicles' class is one of the end objectives, handling glint is an important topic [Fig. 2(c) and (d)].
3) *Shadows:* High-rise structures (trees and buildings) often cast shadows that act as natural occlusions in scene understanding. Fig. 2(f) shows an image where a car is stationed right beside a building but is nearly invisible to the human eye.

Conversion Into Reflectance Data: We calculate the surface reflectance from the calibrated radiance image using the software, ENVI. Calibration panels were deployed in the scene during the various overpasses (Fig. 3). The reflectance

[3]Data analyses were done using ENVI version 4.8.2 (Exelis Visual Information Solutions, Boulder, CO, USA).

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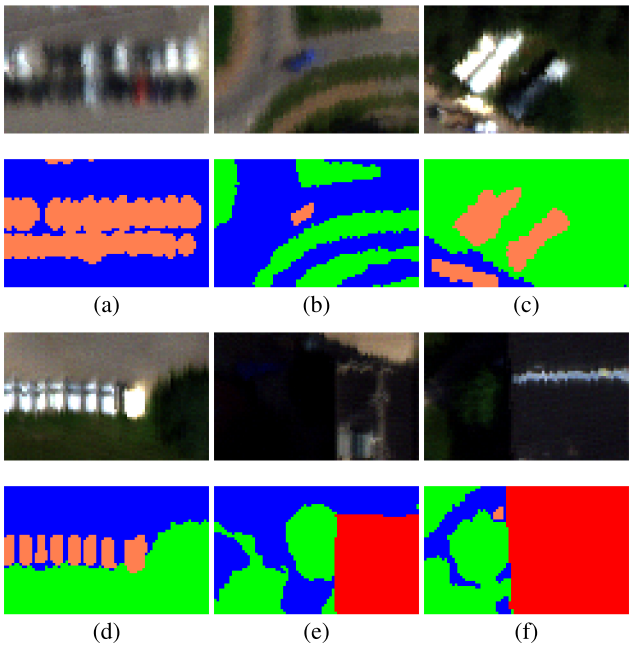


Fig. 2. Challenges present in the AeroRIT scene. (a) and (b) Low resolution. (c) and (d) Glint. (e) and (f) Shadow. Each figure shows the RGB-visualized hyperspectral chip and its corresponding semantic map.



Fig. 3. Targets (cyan) placed in the scene as calibration panels. We use the ground versus aerial signatures to draw a linear mapping between radiance and corresponding reflectance units.

of these black and white uniform calibration panels was measured using a field-deployable point spectrometer. The panels were large enough to produce full pixels in the image data (i.e., minimal pixel mixing). These full pixels enabled us to produce a linear spectral (i.e., per-band) lookup table (LUT) for the mapping of radiance to reflectance. That is, an LUT is generated for every band. This in-scene technique is often called the empirical line method (ELM). One of the key assumptions with this technique is that the atmospheric mapping of radiance into reflectance over the in-scene panels used to define the mapping also applies, spatially, to the rest of the image. This assumption holds fairly true for our case as the atmosphere, spatially, throughout the scene was fairly invariant and uniform. Furthermore, the risk of multiple scattering (i.e., a nonlinear issue) was very minimal due to the fact that the atmospheric conditions were so clear.

IV. CNN-Based Network Architectures

We discuss the encoder–decoder-based CNN architectures that are used to establish benchmarks on the AeroRIT data set.

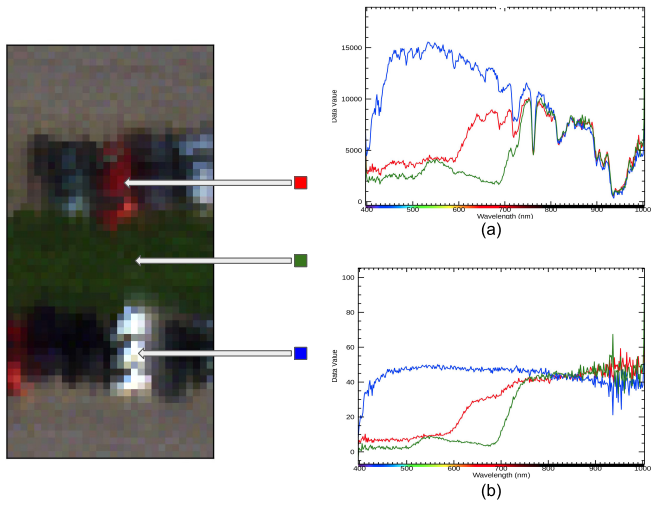


Fig. 4. Comparison of signals obtained from the radiance and reflectance domains. As seen, radiance-a has a varying range of amplitudes while reflectance-b is restricted to the 0%–100% range. The x-axis on the graph denote the bands, and the y-axis on the graph denote the value.

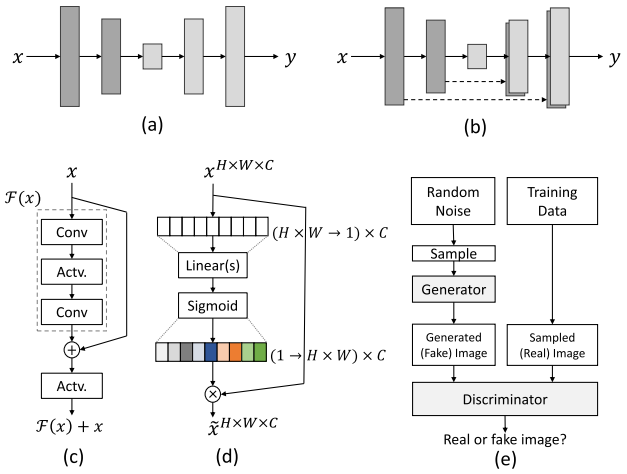


Fig. 5. Graphic description of some models used in this article. (a) SegNet [4]. (b) U-Net [3]. (c) Residual layer [23] connection in Res-U-Net. (d) Squeeze-and-excitation layer from SENet [24]. (e) Workflow of GANs [25]. Generally, the x in (a) and (b) is the image input, and y is the mapping to be learned, e.g., depth estimation, semantic segmentation, and colorization.

With respect to hyperspectral imagery, the model architectures are constrained by the following requirements: 1) they should be able to process low-resolution features very well due to the nature of the data; 2) they should be able to propagate information to all layers of the network so that valuable information is not lost during sampling operations; 3) they should be able to make the most out of limited data samples; and 4) they should be as lightweight with respect to parameters as possible as the data itself are too large in size. The natural choice of selection would be a U-Net [Fig. 5(b)], as the skip connections help propagate additional information from the encoder to the decoder. In technical terms, each skip connection concatenates all channels at layer i with those at layer $n-i$, where n is the total number of layers. We consider SegNet, U-Net, and Res-U-Net (Fig. 5) in our initial studies. As there are no pretrained models available for image

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processing in the hyperspectral domain, we train the networks from scratch. Furthermore, we also investigate two additional approaches that have shown to work in RGB domain: 1) SE block and 2) GANs. The former is used for improving channel interdependences in the network, and the latter is used for self-supervised representation learning in some cases. We further discuss the two approaches in the subsequent sections.

A. SE Block

This layer block [Fig. 5(d)] was proposed by Hu *et al.* [24] to scale network responses by modeling channelwise attention weights. This is similar to a residual layer [Fig. 5(c)] used in ResNets, except that the latter focuses on spatial information as compared to channel information. The workings of this layer are as follows. For any given feature block x, it is passed through global average pooling to obtain a channel feature vector, which embeds the distribution of channelwise feature responses (1). This is referred to as the **squeeze** block. The vector z in $\mathbb{R}^C$ (where C is the number of channels) is generated by squeezing x through its spatial dimensions $H \times W$, such that the cth element of $\mathbf{z}$ is calculated by

$$z_c = \mathbf{F}_{\text{squeeze}}(\mathbf{x}_c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c(i, j). \tag{1}$$

This is followed by two fully connected layers (W_1, W_2) and a sigmoid layer (σ), in which the channel-specific weights can be learned through a self-gating mechanism:

$$\mathbf{s} = \mathbf{F}_{\text{excite}}(\mathbf{z}, W_1, W_2) = \sigma(W_2 \delta(W_1 \mathbf{z})) \tag{2}$$

where δ refers to the ReLU nonlinearity [27], $W_1 \in \mathbb{R}^{\frac{C}{r} \times C}$, $W_2 \in \mathbb{R}^{C \times \frac{C}{r}}$, and r is the reduction ratio to vary the capacity of the block. This is referred to as the *excitation* block. The output of the SE block is obtained by reshaping the learned channel weights (2) to the original spatial resolution and multiplying with the feature block

$$\widetilde{\mathbf{x}_c} = s_c \, \mathbf{x}_c. \tag{3}$$

The final representation $\widetilde{\mathbf{x}}$ is the combination of all $\widetilde{\mathbf{x}_c}$ (3) and provides a more effective channel-weighted feature map that can be passed to the next set of layers.

B. Conditional Generative Adversarial Networks (cGANs)

cGANs were first proposed by Mirza and Osindero [28] and have been widely used for generating realistic-looking synthetic images [5], [29]–[31]. We first discuss the base GAN and then proceed to cGANs framework. A typical GAN [Fig. 5(e)] consists of a generator (G) and a discriminator (D), both modeled by CNNs, tasked with learning meaningful representations to surpass each other. The generator learns to generate new *fake* data instances (e.g., images and audio signals) that cannot be distinguished from the *real* instances, while the discriminator learns to evaluate whether each instance belongs to the actual training data set or is fake/synthetic (created by the generator). Formally, we can write the objective loss function as follows:

$$\mathcal{L}_{\text{GAN}}(G, D) = \mathbb{E}_y[\log D(y)] + \mathbb{E}_z[\log(1 - D(G(z))] \tag{4}$$

where the input to G is sampled from a noise distribution z (e.g., normal, uniform, and spherical) and $\mathbb{E}_y$ is the expectation over the sample distribution (in this case, y). The generator learns a mapping $G : z \to y$ and tries to minimize the loss, while the discriminator tries to maximize it.

In a cGAN setting, the input to generator is no longer just from a noise distribution, but instead appended with a source label x. It now learns a mapping $G : \{x, z\} \to y$, and the corresponding loss function becomes

$$\mathcal{L}_{\text{cGAN}}(G, D) = \mathbb{E}_{x,y}[\log D(x, y)] + \mathbb{E}_{x,z}[\log(1 - D(x, G(x, z))]. \tag{5}$$

Equation (5) shows us that the source label x is also passed on the discriminator, which uses this additional information to perform the same task as in GAN. We use the cGAN-based image to image translation framework of Isola *et al.* [5], with the final objective of the generator as follows:

$$G^* = \arg \min_G \max_D \mathcal{L}_{\text{cGAN}}(G, D) + \lambda \mathcal{L}_{\text{other}}(G) \tag{6}$$

that is, generate samples of a quality that lowers the discriminator's ability to identify if the sample is from the real or fake distribution. The other loss in (6) is an additional term imposed on the generator, which forces the generated image to be as close to the ground truth as possible. We use the standard L1-loss as $\mathcal{L}_{\text{other}}(G)$.

Self-supervised learning (survey: [32]) has shown much potential in helping randomly initialized neural networks learn better initialization points before being applied for their original task in other domains. We apply image in-painting and image denoising as two tasks for self-supervision on our data set. The two tasks can be described as follows: 1) in-painting: randomly generate binary masks and multiply them with the real image and 2) image denoising: perturb the original image with Gaussian or salt-and-pepper noise. The networks are then tasked with restoring the original image from the corrupted image. Obtaining a good quality representation of the underlying pixels, in turn, helps the network learn a weak prior over the image space. We adopt the above discussed cGAN framework and experiment with both the tasks. The entire training framework is summarized in Fig. 8.

V. Experiments and Results

A. Experimental Configurations

We use the PyTorch library [33] for all our experiments. We split the scene into training, validation, and test as follows: the original flight line was 1973×3975. We drop the first 53 rows and 7 columns and get a flight line of 1920×3968. We use the first 1728 columns (and all rows) for training, the next 512 columns as validation and the last 1728 as the test split. We sample 64×64 patches (with 50% overlap) to create a training set and nonoverlapping patches for validation

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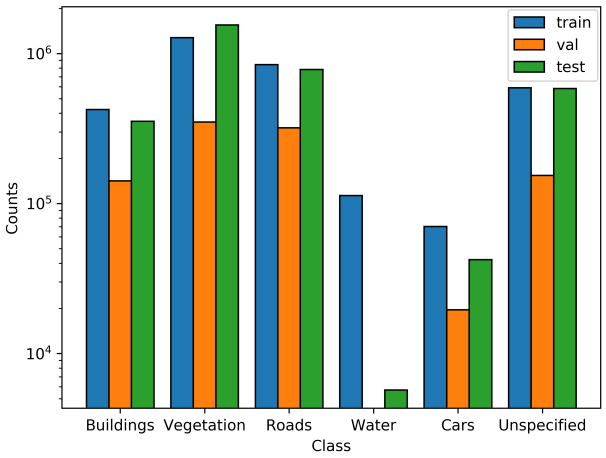


Fig. 6. Label distribution of the data set in log space. Cars are comparatively under-represented in the scene, while vegetation and roads have the highest number of samples.

and test sets. Fig. 6 shows the number of samples present in each class—the scene is heavily imbalanced with reference to class *cars*. We adopt basic data augmentation techniques, random flip, and rotation, and extend the data set by a factor of four. We use a batch size of 100 and train for 60 epochs with a learning rate of 1e-4. We also use a multistep decay of factor 0.1 at epochs 40 and 50.

We sample every tenth band from 400 to 900 nm (i.e., 400, 410, . . . , 900 nm) to obtain 51 bands from the entire band range. As the 372 band centers are not aligned in perfect order, we use ENVI for extracting near accurate band centers. In preliminary experiments, we found that the last set of bands (from 90 to 1000 nm) did not provide useful discriminative information (intuitively due to the low signal-to-noise ratio in the channels), so they were removed for all experiments. We normalize all data between 0 and 1 by clipping to a max value of 2^{14} (16 384).

B. Loss and Metrics

We use weighted categorical cross-entropy to minimize the segmentation map and ignore the *unspecified* class label. The weights are calculated using median-frequency balancing [34], where the number of pixels in the scene belonging to a particular class is also taken into consideration. This helps overcome the class imbalance shown in Fig. 6.

We use the following sets of metrics: overall accuracy (OA), mean per-class accuracy (MPCA), mean Jaccard index (popularly known as mIOU), and mean Sørensen–Dice coefficient (mDICE).

OA and MPCA report the percentage of pixels correctly classified. However, they are still slightly prone to a data set bias when class representation is small, and hence, we also report mIOU and mDICE. mIOU is the classwise mean of the area of intersection between the predicted segmentation and the ground truth divided by the area of union between the predicted segmentation and the ground truth. Correlated with mIOU, mDICE also focuses on intersection over union and is often used as a secondary metric for measuring a network's performance on the task of semantic segmentation. We adopt mIOU as the primary metric for the measure of performance.

TABLE II
PERFORMANCE OF VARIOUS MODELS USED FOR ESTABLISHING BASELINE ON THE AERORIT TEST SET

	pixel acc. (OA)	mean pixel acc. (MPCA)	mIOU	mDICE
SegNet	92.12	72.97	52.60	61.50
U-Net	93.15	72.90	60.40	68.63
Res-U-Net (6)	93.28	88.09	72.55	82.15
Res-U-Net (9)	93.25	84.64	70.88	80.88
SegNet-m	93.20	74.86	59.08	67.41
U-Net-m	93.25	89.66	70.62	80.86
U-Net-m (ours)	**93.61**	**90.67**	**76.40**	**85.60**

TABLE III
IMPACT OF EACH COMPONENT ADDED TO THE BASELINE U-NET-M MODEL FROM TABLE II

SE layer	SE act. PReLU	cGAN	mean pixel acc. (MPCA)	mIOU	mDICE
			89.66	70.62	80.86
✓			88.59	75.35	84.05
✓	✓		90.28	75.89	84.81
✓	✓	✓	**90.67**	**76.40**	**85.60**

C. Model Hyperparameters

SegNet and U-Net both have encoders with four max-pooling layers and gradually increasing channels by power of 2 ($C64 - \text{MP} - C128 - \text{MP} - C256 - \text{MP} - C512 - \text{MP} - \text{BottleNeck}$, where C is the number of channels and MP indicates a max-pooling operation). Res-U-Net blocks are built upon U-Net, and conventionally, Res-U-Net (N) contains N identity mapping residual blocks for better information passing. In our experiments, we use $N = 6$ and $N = 9$ following [5]. We also use smaller versions of SegNet and U-Net, called *SegNet-m, U-Net-m*, which drop the number of max-pooling layers from 4 to 2 to compensate for the scene's low spatial resolution and increase the channel by a factor of 2.

D. Results

We compare all the models trained for the task of semantic segmentation in Table II. We observe that 6-block Res-U-Net achieves the best performance, but as U-Net-m has nearly four times fewer parameters and roughly the same performance, we adopt U-Net-m as the baseline in this article. We develop on this baseline and achieve a better performing U-Net-m version that outperforms all previous baselines.

We discuss the approaches used to further improve the performance of the U-Net-m architecture (Table III and Fig. 7). We adopt the inception-variant of SE block with a reduction ratio $r = 2$—we do not sum the output of the SE block with

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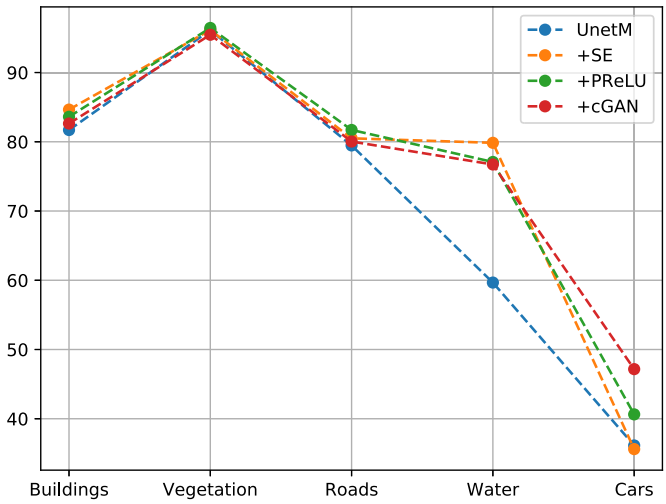


Fig. 7. Results with various additions to normal U-Net-m. The y-axis is the IOU measure. SE-block and its additions improve the performance of water pixel identification by nearly 20 points over the baseline, and the overall modifications improve the performance of car pixel identification by eight points. Self-supervised learning is the factor that contributes to the large improvement in car pixel identification.

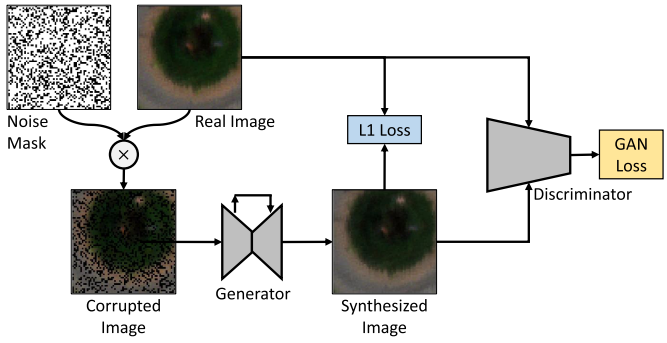


Fig. 8. Procedure for image reconstruction from a corrupted image. The generator is the network under consideration (U-Net-m-SE-PReLU), and the discriminator has five convolution layers followed by batch normalization and leaky ReLU.

the original channel space as a skip connection, but use it as the importance-weighted channel output [24]. We add an SE block after every $conv - batchnorm - relu$ combination on the encoder side of the network. This increases the U-Net-m performance by almost four points. We further replace every ReLU activation with parameterized ReLU (PReLU) and observe a slight performance boost [26].

We apply image in-painting and image denoising as the two tasks for self-supervision, and we found in-painting to work as a better technique in the preliminary experiments. Once the network is trained on in-painting, we then retain the encoder weights and train the decoder from scratch on semantic segmentation. This approach further increases the performance by one point and has the cleanest inference labels (Fig. 9).

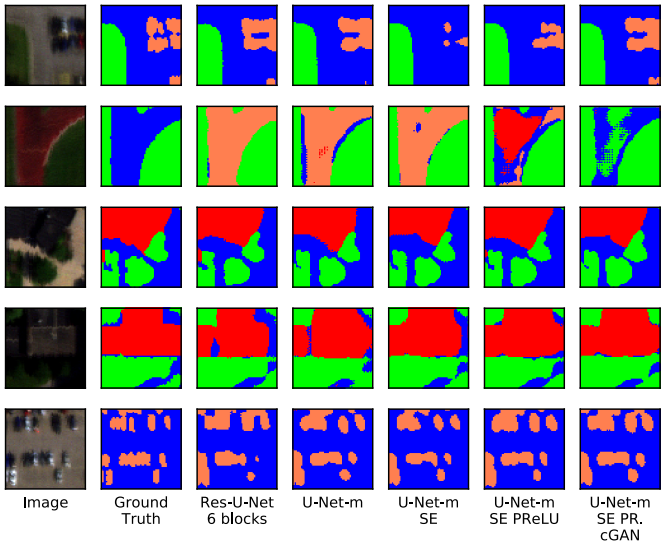


Fig. 9. Successful cases: outputs for a set of images among all networks. The racetrack image (row 2) shows that the cGAN trained network is the only one that is able to understand that the red unseen track patch is not a car or building.

VI. Discussion

Along with the promising performance that CNNs have achieved in the hyperspectral domain [18]–[20], [35], there are several directions of research that can be construed as open problems on this data set.

A. Radiance or Reflectance

We compared the performance of U-Net-m-SE on the radiance and reflectance sets of images and obtain an mIOU of nearly five points less when using reflectance (reflectance-mIOU: 69.90 versus radiance-mIOU: 75.35). We hypothesize that the difference in discriminative signatures (Fig. 4) might be one of the reasons behind the performance loss. As reflectance is the atmosphere rectified version and is theoretically less noisy, the performance drop is intriguing. However, as our primary domain of interest is the radiance space, we do not investigate this further and leave analyzing reflectance as an open problem.

B. Data Augmentation

We identify some of the tricky cases within the flight line in Fig. 10. Rows 1–3 show images partly under the shadow that have misclassified pixels, row 4 shows a pedestrian crossing misclassified as a vehicle, and row 5 shows an object shining inside the fountain due to glint. The signatures vary heavily in amplitude due to shadows and glint and, hence, cause the networks to confuse between different classes. Conventional RGB augmentations (brightness and contrast) rely on the assumption of uniform scene illumination irrespective of image size. However, as hyperspectral signatures can vary drastically under varying atmospheric conditions and are subject to the adjacency effect, it is not possible to directly apply RGB-based augmentations for improving performance.

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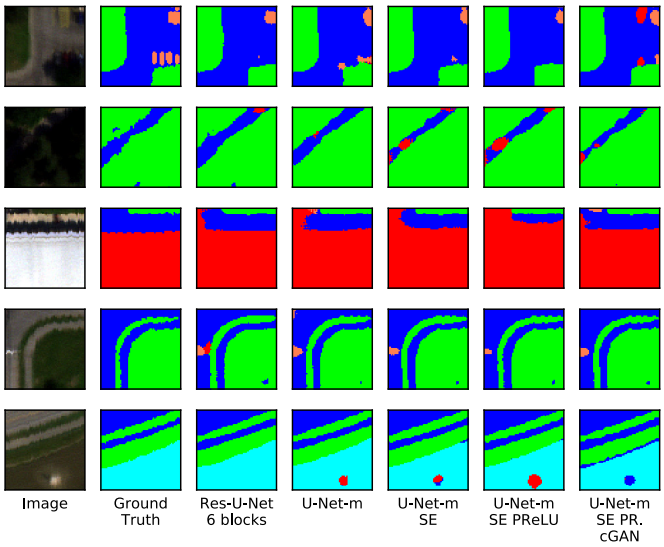


Fig. 10. Failure cases: outputs for a set of images through all trained networks. All networks predict building in between the road in row 2 and misclassify zebra crossing as a car.

C. Understanding the Workings of CNNs

Learning task-relevant representations has been the forte of CNNs, and a lot of techniques have been proposed to understand their internal workings [36]–[38]. However, none of these techniques have been applied toward hyperspectral imagery, and CNN architectures are often treated as black-box approximators giving constant performance improvements. This approach is not favorable—knowing why a particular pixel has been classified as belonging to "cars" or why limiting max-pooling layers to 2 (and indirectly, the receptive field) boosted the performance can, in turn, help design better architectures. Hence, there is a need to understand the why and how of CNNs with respect to HSI: more importantly, to analyze how much of the pixels' classification depends on its spatial information (that is, structure) as compared to the spectral information (spectrum).

VII. Conclusion

This article introduces AeroRIT, the first large-scale aerial hyperspectral scene with pixelwise annotations. Our scene is nearly eight times bigger than the previously largest scene and is composed of challenging factors such as shadows, glint, and mixed pixels that make inference difficult. We trained networks for semantic segmentation and established a baseline using SE block, self-supervision, and PReLU activation. We believe AeroRIT can be used for future work in multiple areas of remote sensing, but not limited to data augmentation and network architecture designing.

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